ECN 594: Introduction and Foundations

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Plan for today

1. **What is industrial organization?**
2. Course structure and logistics
3. Why do we need demand models?
4. Ordinal vs cardinal utility
5. Random utility framework
6. The logit model
7. Berry (1994) inversion

Motivation

- Consider two forms of competition: **perfect competition** and **monopoly**.
- We can think of these two forms of competition as polar opposites.
- **Perfect competition** (supply and demand)
  - Many tiny 'atomistic' firms
  - Total surplus is maximized and so the market is 'efficient'
- **Monopoly**
  - One single firm
  - Monopoly sets price 'too high'; the market is inefficient

Most real-world markets do not fit neatly into these two categories

- Some examples in this course:
  1. Firms set different prices for the same good (airline tickets, student discounts)
  2. Only a few big firms in the market (health insurance, tech)
  3. Firms collude to raise prices (OPEC, cartels)
  4. Firms merge with other firms (subject to **antitrust** laws)

What is industrial organization (IO)?

- **IO is the study of firm and consumer behavior in markets between (and including) the polar opposites of perfect competition and monopoly.**
- Why is this useful?
  1. <u>Designing regulation/antitrust policy</u>:
     - When do markets fail? How should we regulate?
  2. <u>Firm strategy</u>:
     - How to set prices? Design marketplaces?

IO is central to many policy debates right now

- Should we break up Google, Amazon, Meta?
- Are hospital mergers raising healthcare costs?
- Can algorithms learn to collude?

These are all IO questions.

This course has two components

- **Theoretical:** Models of pricing, oligopoly, entry, mergers, vertical relationships
- **Empirical:** Demand estimation, merger simulation
- The empirical content (demand estimation) is typically taught at the PhD level
- But it's so useful in practice that we cover it here
- You'll estimate demand using Python and the `pyblp` package

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Course structure

1. **Part 1 (Lectures 1-7):** Demand Estimation and Pricing
   - Random utility models, logit demand
   - Identification, instrumental variables
   - Estimation with `pyblp`
   - Price discrimination
2. **Part 2 (Lectures 8-14):** Competition and Industry Structure
   - Cournot, Bertrand, Hotelling
   - Entry, mergers, collusion
   - Vertical relationships

Logistics

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- Office: CRTVC 455G
- Office Hours: By appoinment (email me)
- **Textbook:** Cabral, *Introduction to Industrial Organization* (2nd ed.)
- For demand estimation: selected readings (posted on Canvas).

Assessment

- **20%** Homework 1
- **20%** Homework 2
- **30%** Midterm (Feb 9)
- **30%** Final (Mar 4)
- Exams: 80 minutes, calculator + one two-sided cheat sheet allowed

Building on your prior courses

- From **ECN 565** (Alvin Murphy): Discrete choice econometrics
  - You know logit, probit, MLE
  - We'll apply this to IO problems
- From **ECN 532** (Hector Chade): Game theory
  - You know Nash equilibrium, Cournot, Bertrand, repeated games
  - We'll do quick refreshers and focus on IO applications

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Why is estimating demand useful?

- You may be surprised why we are starting with demand estimation, given that I motivated IO as about firm behavior
- But it turns out that **estimating demand is central to many IO questions**, including:
  - Setting optimal prices
  - Effects of a merger on prices
  - Value of new goods
  - Any question about consumer welfare

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Ordinal vs cardinal utility

- **Ordinal utility:** only rankings matter, not magnitudes
  - $U(A) > U(B)$ means I prefer A to B
  - But "$U(A) = 2 \times U(B)$" has no meaning
- This is the standard microeconomics assumption
- **Problem:** We want to measure consumer surplus in $!
  - "How much better off are consumers from this policy?"
  - Need utility to have a meaningful *scale*

Quasi-linear utility makes utility cardinal

- **Solution:** Assume quasi-linear utility

$$U = u(\text{goods}) + y$$

  where y is income (or "money left over")

- Take the derivative with respect to income:

$$\frac{\partial U}{\partial y} = 1$$

- The **marginal utility of income is constant** (and equals 1)

- This means we can measure utility in dollars!

Why this matters

- With quasi-linear utility:
  - 1 util = \$1
  - Consumer surplus has a meaningful interpretation
  - We can add up utility across consumers
- This is the standard assumption in IO demand estimation
- (In contrast to general equilibrium models where income effects matter)

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Random utility framework (refresher from Alvin's course)

- Consumer i chooses among J products
- Utility of consumer i for product j:

$$u_{ij} = V_{ij} + \varepsilon_{ij}$$

- V_{ij}: deterministic component (observed by econometrician)
- ε_{ij}: random component (taste shock)

Random utility framework

- Consumer i chooses product j if:

$$u_{ij} > u_{ik} \quad \text{for all } k \neq j$$

- Choice probability:

$$P(i \text{ chooses } j) = P(u_{ij} > u_{ik} \text{ for all } k)$$

- Different assumptions on ε_{ij} give different models:
  - Type I Extreme Value $\rightarrow$ **Logit**
  - Normal $\rightarrow$ **Probit**
- You covered this in ECN 565; we'll apply it to IO

Discrete choice in IO

- In IO, we typically write:

$$u_{ijt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

- i: consumer, j: product, t: market
- x_{jt}: observed product characteristics (size, horsepower, etc.)
- p_{jt}: price
- ξ_{jt}: unobserved (by the econometrician) product quality
- ε_{ijt}: idiosyncratic taste shock (also unobserved)
- $\alpha < 0$: price coefficient (higher prices reduce utility)

What is a "market"?

- We index products by market t. What defines a market?
- **Geography:**
  - City? State? Country?
  - Depends on whether products/prices vary across locations
- **Time:**
  - Week? Month? Quarter? Year?
  - Depends on frequency of price changes, data availability
- **Examples:**
  - Cars: national market, annual (BLP 1995)
  - Cereal: city-quarter (Nevo 2001)
  - Airlines: origin-destination route, quarterly

What is ξ_{jt}: more detail

- ξ_{jt} = unobserved (by the econometrician) product quality
- Examples (in the car market):

What is ξ_{jt}: more detail

- ξ_{jt} = unobserved (by the econometrician) product quality
- Examples (in the car market):
  - Brand equity ("I just like Toyota")
  - Advertising effects
  - Design/style
  - Reputation
- Key insight: firms *observe* ξ_{jt} when setting prices!
- This creates an endogeneity problem (more on this next lecture)

Why isn't income in the utility function?

- You might expect: $u_{ijt} = x_{jt}\beta + \tilde{\alpha}(y_i - p_{jt}) + \xi_{jt} + \varepsilon_{ijt}$

- But we write: $u_{ijt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} + \varepsilon_{ijt} \quad (\alpha < 0)$

- **Why?** (think about what you know about discrete choice models from Alvin's course)

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- But we write: $u_{ijt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} + \varepsilon_{ijt} \quad (\alpha < 0)$
- **Why?** (think about what you know about discrete choice models from Alvin's course)
- With quasi-linear utility, income enters linearly
- When comparing alternatives, income *cancels out*:

$$\begin{aligned} u_{ijt} - u_{ikt} &= [x_{jt}\beta + \tilde{\alpha}(y_i - p_{jt}) + \xi_{jt} + \varepsilon_{ijt}] \\ &\quad - [x_{kt}\beta + \tilde{\alpha}(y_i - p_{kt}) + \xi_{kt} + \varepsilon_{ikt}] \\ &= (x_{jt} - x_{kt})\beta + \alpha(p_{jt} - p_{kt}) + (\xi_{jt} - \xi_{kt}) + (\varepsilon_{ijt} - \varepsilon_{ikt}) \end{aligned}$$

- Only *differences* matter for choice, and y_i drops out

The outside option

- Important: Consumers can choose **not to buy** any product
- This is the **outside option** (product 0)
- Utility of outside option:

$$u_{i0t} = \varepsilon_{i0t}$$

- We normalize non-idiosyncratic components to 0
- All other utilities are *relative* to the outside option
- Why this matters:
    - If prices rise, consumers can “exit” the market
    - This affects elasticities and market power

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The logit assumption

- Random utility: $u_{ijt} = \delta_{jt} + \varepsilon_{ijt}$ where $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} \quad (\alpha < 0)$
- What distribution should ε_{ijt} have?
- Assume ε_{ijt} is i.i.d. **Type I Extreme Value** (Gumbel)
- CDF: $F(\varepsilon) = \exp(-\exp(-\varepsilon))$
- Why this assumption?
  - Gives us **closed-form** choice probabilities
  - Computationally tractable

Logit choice probabilities

- With Type I Extreme Value errors, the probability that consumer chooses j:

$$P(\text{choose } j) = \frac{\exp(\delta_{jt})}{\sum_{k=0}^{J} \exp(\delta_{kt})}$$

- This is the **logit** formula (you've seen this in ECN 565)

- Including the outside option ($\delta_{0t} = 0$):

$$s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}, \qquad s_{0t} = \frac{1}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}$$

- Note: $s_{0t} + \sum_{j=1}^{J} s_{jt} = 1$ (shares sum to 1)

Why logit is useful

- **Closed form:** No numerical integration needed
- **Invertible:** Can go from shares to mean utilities (Berry inversion)
- **Elasticities:** Simple formulas for own- and cross-price elasticities

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Berry (1994) inversion: the key insight

- We observe: market shares s_{jt}
- We want: mean utilities δ_{jt} (to estimate β, α)
- **Problem:** How do we get δ_{jt} from s_{jt}?
- **Berry's insight:** Take the log of shares!

Berry (1994) inversion

- Start with:

$$s_{jt} = \frac{\exp(\delta_{jt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}, \quad s_{0t} = \frac{1}{1 + \sum_{k=1}^{J} \exp(\delta_{kt})}$$

- Take logs:

$$\ln(s_{jt}) = \delta_{jt} - \ln\left(1 + \sum_{k=1}^{J} \exp(\delta_{kt})\right)$$
$$\ln(s_{0t}) = -\ln\left(1 + \sum_{k=1}^{J} \exp(\delta_{kt})\right)$$

- Subtract:

$$\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$$

Berry (1994) inversion: the estimating equation

- We have: $\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$
- Substitute $\delta_{jt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$ (where $\alpha < 0$):

$$\ln(s_{jt}) - \ln(s_{0t}) = x_{jt}\beta + \alpha p_{jt} + \xi_{jt}$$

- This is a **linear regression**!
- LHS: can compute from observed shares
- RHS: product characteristics, price, and an error term

Worked example: Berry inversion

- **Question:**
- Market has 2 products plus outside option. Observed shares:

$$s_{0t} = 0.5, \quad s_1 = 0.3, \quad s_2 = 0.2$$

- Compute the mean utilities δ_1 and δ_2.

Take 2 minutes to solve this.

Worked example: Berry inversion (solution)

- **Solution:**
- Berry inversion: $\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$
- For product 1:

$$\delta_1 = \ln(0.3) - \ln(0.5) = \ln(0.3/0.5) = \ln(0.6) \approx -0.51$$

- For product 2:

$$\delta_2 = \ln(0.2) - \ln(0.5) = \ln(0.2/0.5) = \ln(0.4) \approx -0.92$$

- Interpretation: Both products have negative mean utility relative to outside option
- Product 1 is "better" than product 2 (higher δ)

Key Points

1. **IO** studies firm behavior between perfect competition and monopoly
2. This course has **theory** (pricing, oligopoly) and **empirical** (demand estimation) components
3. **Demand estimation** is central to IO: elasticities $\rightarrow$ market power $\rightarrow$ policy
4. **Quasi-linear utility** makes consumer surplus measurable in \$
5. **Random utility:** $u_{ijt} = x_{jt}\beta + \alpha p_{jt} + \xi_{jt} + \varepsilon_{ijt} \quad (\alpha < 0)$
6. **Logit:** Type I EV errors give closed-form choice probabilities
7. **Berry inversion:** $\ln(s_{jt}) - \ln(s_{0t}) = \delta_{jt}$ turns demand into a regression

Next time

- **Lecture 2:** Identification and Instrumental Variables
    - Elasticity formulas
    - Why price is endogenous
    - Direction of bias
    - Instrumental variables (cost shifters, BLP IVs)